HSLU Hochschule Luzern

AI EVENT · LIVE DEMO


AUTHOR
Elias Martinelli
LinkedIn


AUTHOR
Timo Schlumpf
LinkedIn


SUPERVISOR
Dr. Elena Nazarenko
LinkedIn

DSPRO1 · Team 8 · Spring 2026

DATA / MACHINE LEARNING / REAL ESTATE / SWISS OPEN DATA

PREDICTING APARTMENT RENTAL PRICES IN SWITZERLAND

A supervised machine-learning approach for the Swiss cold-rent market

We built a reproducible pipeline that scrapes Swiss rental listings, enriches them with official building (GWR) and swisstopo location data, and predicts the monthly cold rent of a single apartment. From 9'994 scraped listings, 4'536 fully enriched apartments were used for modelling. GradientBoosting with the ALL+geo feature set is the final model: not the lowest error, but the best balance of accuracy and generalisation.

01 THE PROBLEM

Can a single apartment's rent be judged fairly?

2-room flat · City of Lucerne

WHAT SHOULD IT COST?

Asking rents vary widely for similar flats

Location dependency
The same flat costs far more in the city of Lucerne than in rural regions.

Many interacting features
Area, rooms, building age and local market interact non-linearly.

No apartment-level benchmark
Official market indicators do not price one specific flat.

02 FROM DATA TO MODEL

From 25'002 listing URLs to 4'536 enriched apartments.

25'002 URLs identified
rentumo.ch, one scrape on 13 Apr 2026

9'994 listings scraped
detail pages successfully extracted

9'582 with core fields
address · area · rooms · cold rent

4'536 fully enriched
GWR building + swisstopo location data

ENRICHMENT STACK

rentumo.ch · GeoAdmin · GWR / BFS · swisstopo

MODEL INPUT · 11 PREDICTORS + GEO CLUSTER

Apartment: area, rooms · Building: year built, dwellings, land area · Location: coordinates, elevation, population, public-transport score, solar class

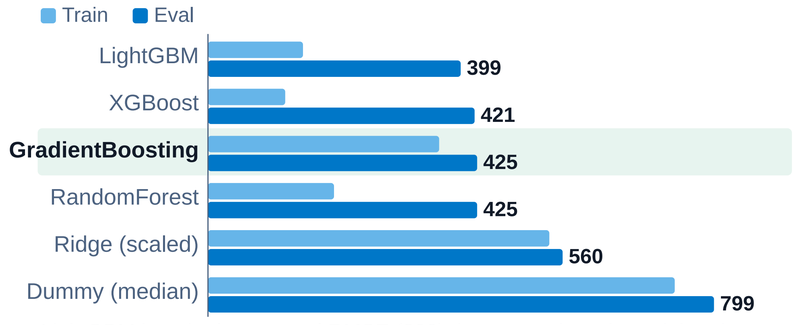
03 RESULTS

Every tree model clearly beats the linear benchmark.

FINAL MODEL
GRADIENTBOOSTING · ALL+GEO

EVAL RMSE	EVAL R^2	TRAIN–EVAL GAP
CHF 425	0.710	CHF 60

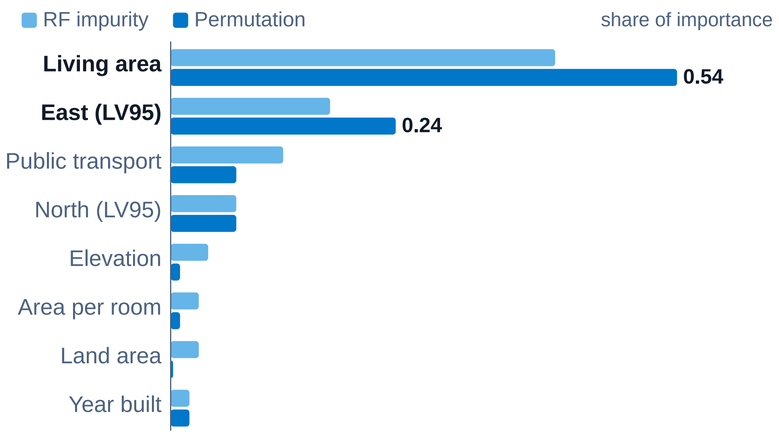
RMSE train vs. eval (CHF, engineered features)



LightGBM has the lowest eval RMSE (399), but memorises the training data (gap 249). Bootstrap intervals of the top tree models overlap, so the smaller gap decides.

04 WHAT DRIVES RENT?

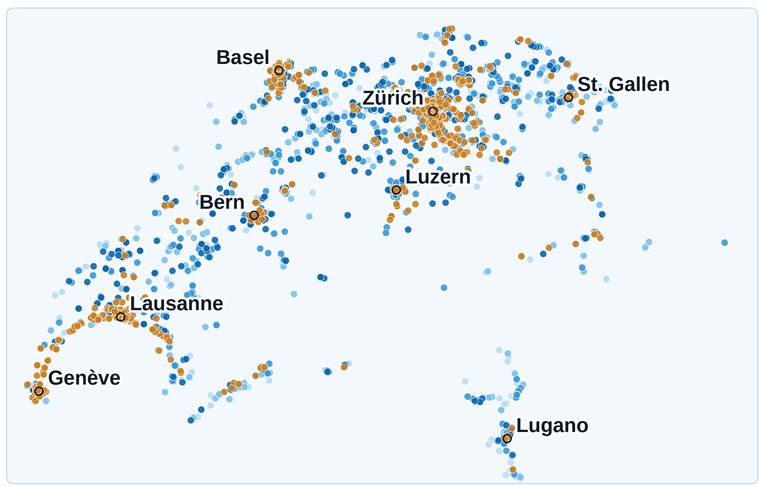
Living area is the strongest single predictor; location comes next.



Size and coordinates dominate every method. Public-transport access and elevation add local-market context.

05 WHERE IS RENT HIGH?

4'536 modelling apartments, coloured by monthly cold rent.



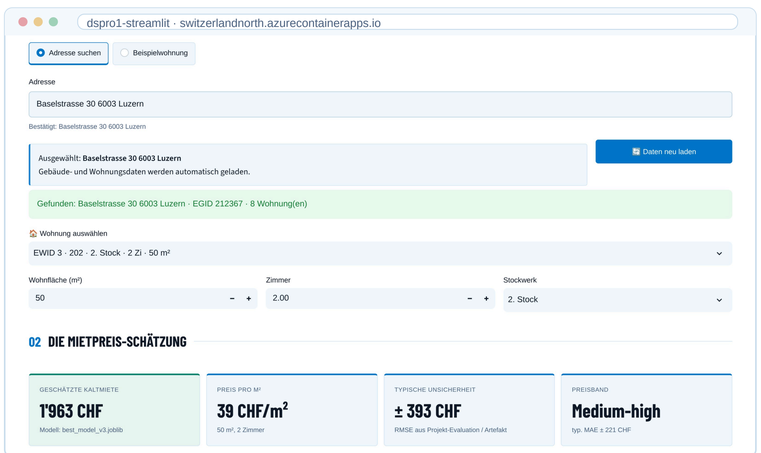
Cold rent, CHF / month

< 1'250 · 1'250–1'550 · 1'550–1'850 · 1'850–2'300 · ≥ 2'300

Zürich, Geneva, Basel and Lausanne form dense, high-rent clusters; urban markets are over-represented in the data.

06 TRY THE PROTOTYPE

Swiss address → official GWR dwelling → live rent estimate.



Live example: Baselstrasse 30, 6003 Luzern · 2nd floor, 50 m², 2 rooms

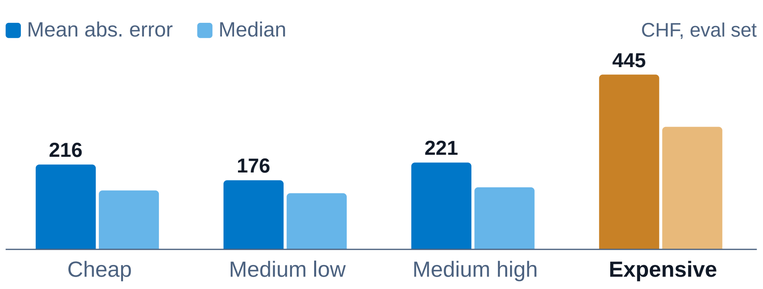
SCAN TO EXPLORE

Estimate a monthly cold rent for any Swiss address.

Demo model: LightGBM + KNN, eval RMSE 393

07 WHERE DOES IT STRUGGLE?

The most expensive quartile has about twice the error.



Missing signals such as view, floor, renovation state and finish matter most for premium flats, mainly around Zürich and Lake Geneva.

08 LIMITATIONS

Snapshot dataset
one scrape on 13 April 2026, no market trend

Coverage bias
urban listings over-represented, rural areas sparse

Missing qualitative signals
view, floor, renovation, balcony, furnishing

High-price uncertainty
largest errors for rare luxury apartments

Overlapping model CIs
the final choice is a robustness decision

09 NEXT STEPS

Features from text + images
extract floor, view and condition

More listing platforms
e.g. Homegate, ImmoScout24

Repeated snapshots
monthly scrapes instead of one

Time-aware validation
retrain and test on newer data

Production deployment
model versioning and drift monitoring

ACCURACY IS NOT EVERYTHING.

The final model was chosen because it balances accuracy, stability and explainability.

Code: github.com/wadafacc/dspro1

Data: rentumo.ch · GWR (BFS) · swisstopo / GeoAdmin